Holomorphic Reflexivity with Reductive Identity Component: A Partial Solution to Conjecture 1.5 of arXiv:2002.03617

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Status. Substantial partial result, likely valid, human review needed. The conjecture is proved when the identity component is connected linearly complex reductive, without assuming that the component group is finite.

1 Source problem

For a complex Lie group G, Aristov defines the Banach-algebra linearizer $\widetilde{\mathrm{Lin}}_{\mathbb{C}}(G)$ as the intersection of the kernels of all holomorphic homomorphisms from G into unit groups of Banach algebras. Conjecture 1.5 of [2] asks whether, for every compactly generated Stein group G,

$$\widetilde{\mathrm{Lin}}_{\mathbb{C}}(G) = \{1\} \iff \mathcal{O}(G) \text{ is holomorphically reflexive.}$$

The source proves necessity and proves sufficiency when G is abelian, discrete, or has finitely many components.

2 Result

A connected *linearly complex reductive* group is, equivalently for the present purpose, the universal complexification of a compact Lie group. It is an affine algebraic complex group.

Theorem 1 (Algebraic-restriction criterion)**.** *Let G be a compactly generated complex Lie group and let $H = G_0$ be its connected identity component. Assume that H is linear. Let*

$$r : \mathcal{O}_{\exp}(G) \longrightarrow \mathcal{O}_{\exp}(H)$$

be restriction, and set

$$A_{\mathrm{alg}} := r(\mathcal{O}_{\exp}(G)) \cap \mathcal{R}(H).$$

If

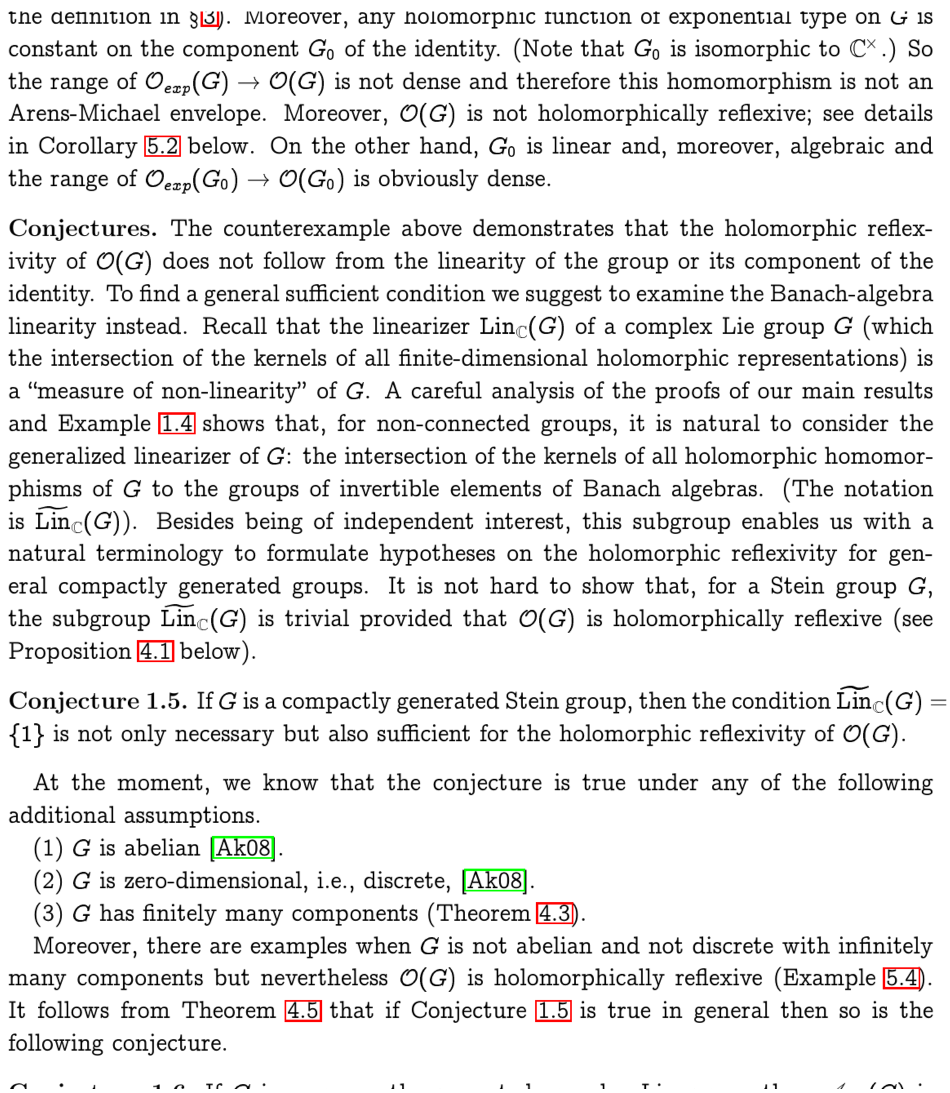
$$N(A_{\mathrm{alg}}) := \{h \in H : a(h) = a(1) \text{ for every } a \in A_{\mathrm{alg}}\} = \{1\},$$

then $\mathcal{O}(G)$ is holomorphically reflexive.

Corollary 2 (Reductive identity component)**.** *Let G be a compactly generated complex Lie group whose identity component G_0 is connected linearly complex reductive. Then*

$$\mathcal{O}(G) \text{ is holomorphically reflexive} \iff \widetilde{\mathrm{Lin}}_{\mathbb{C}}(G) = \{1\}.$$

In particular, Conjecture 1.5 holds in this class even when G/G_0 is infinite.



the definition in §3). Moreover, any holomorphic function of exponential type on G is constant on the component G_0 of the identity. (Note that G_0 is isomorphic to $\mathbb{C}^\times$.) So the range of $\mathcal{O}_{exp}(G) \to \mathcal{O}(G)$ is not dense and therefore this homomorphism is not an Arens-Michael envelope. Moreover, $\mathcal{O}(G)$ is not holomorphically reflexive; see details in Corollary 5.2 below. On the other hand, G_0 is linear and, moreover, algebraic and the range of $\mathcal{O}_{exp}(G_0) \to \mathcal{O}(G_0)$ is obviously dense.

Conjectures. The counterexample above demonstrates that the holomorphic reflexivity of $\mathcal{O}(G)$ does not follow from the linearity of the group or its component of the identity. To find a general sufficient condition we suggest to examine the Banach-algebra linearity instead. Recall that the linearizer $\mathrm{Lin}_{\mathbb{C}}(G)$ of a complex Lie group G (which the intersection of the kernels of all finite-dimensional holomorphic representations) is a "measure of non-linearity" of G. A careful analysis of the proofs of our main results and Example 1.4 shows that, for non-connected groups, it is natural to consider the generalized linearizer of G: the intersection of the kernels of all holomorphic homomorphisms of G to the groups of invertible elements of Banach algebras. (The notation is $\widetilde{\mathrm{Lin}}_{\mathbb{C}}(G)$). Besides being of independent interest, this subgroup enables us with a natural terminology to formulate hypotheses on the holomorphic reflexivity for general compactly generated groups. It is not hard to show that, for a Stein group G, the subgroup $\widetilde{\mathrm{Lin}}_{\mathbb{C}}(G)$ is trivial provided that $\mathcal{O}(G)$ is holomorphically reflexive (see Proposition 4.1 below).

Conjecture 1.5. *If G is a compactly generated Stein group, then the condition $\widetilde{\mathrm{Lin}}_{\mathbb{C}}(G) = \{1\}$ is not only necessary but also sufficient for the holomorphic reflexivity of $\mathcal{O}(G)$.*

At the moment, we know that the conjecture is true under any of the following additional assumptions.

(1) G is abelian [Ak08].

(2) G is zero-dimensional, i.e., discrete, [Ak08].

(3) G has finitely many components (Theorem 4.3).

Moreover, there are examples when G is not abelian and not discrete with infinitely many components but nevertheless $\mathcal{O}(G)$ is holomorphically reflexive (Example 5.4). It follows from Theorem 4.5 that if Conjecture 1.5 is true in general then so is the following conjecture.

Figure 1: Conjecture 1.5 and the previously known cases, from arXiv:2002.03617v3, PDF page 7.

3 Proof intuition

The source reduces reflexivity to density of the restriction map $\mathcal{O}_{\exp}(G) \to \mathcal{O}_{\exp}(G_0)$. The point is to turn separation into density.

Regular restrictions form a translation-invariant algebra. Finite translation orbits of regular functions show that this algebra is actually a Hopf subalgebra of the affine coordinate ring. If its common kernel is trivial, Zariski Noetherianity extracts finitely many coefficient coalgebras whose direct sum is a faithful algebraic representation. The coefficients of a faithful representation and its dual generate the entire coordinate ring, so all regular functions occur as restrictions. Regular functions are dense among exponential-type functions on every connected linear group.

When G_0 is reductive, every exponential-type function on G_0 is already regular. Thus the common kernel of the regular restriction algebra is forced inside $\widetilde{\mathrm{Lin}}_{\mathbb{C}}(G)$, and the conjectured hypothesis makes it trivial. The mechanism is unavailable verbatim for a general solvable identity component, where nonregular exponential-type functions can be essential.

4 Algebraic lemmas

All coordinate rings below are over $\mathbb{C}$.

Lemma 3 (Translation invariance gives a Hopf subalgebra)**.** *Let H be an affine algebraic group and let $A \subset \mathcal{R}(H)$ be a unital subalgebra stable under all left and right translations and under inversion. Then A is a Hopf subalgebra of $\mathcal{R}(H)$.*

Proof. Multiplication, unit, counit, and antipode cause no difficulty. It remains to prove $\Delta(A) \subset A \otimes A$ algebraically.

Fix $a \in A$. Since a is regular, the span

$$V = \operatorname{span}\{R_y a : y \in H\}$$

of its right translates is finite dimensional; translation invariance gives $V \subset A$. Choose a basis $v_1, \dots, v_m$ of V. There are regular functions c_i on H such that

$$a(xy) = \sum_{i=1}^{m} v_i(x) c_i(y).$$

Choose points $x_1, \dots, x_m \in H$ for which the matrix $(v_i(x_j))_{j,i}$ is invertible. The displayed identities with $x = x_j$ show that each c_i is a linear combination of the left translates $y \mapsto a(x_j y)$, hence $c_i \in A$. Therefore

$$\Delta a = \sum_{i=1}^{m} v_i \otimes c_i \in A \otimes A.$$

Inversion stability gives antipode stability, completing the proof. $\square$

Lemma 4 (Trivial-kernel Hopf subalgebras)**.** *Let H be an affine algebraic group and $A \subset \mathcal{R}(H)$ a Hopf subalgebra. If*

$$N(A) := \{h \in H : a(h) = a(1) \text{ for all } a \in A\} = \{1\},$$

then $A = \mathcal{R}(H)$.

Proof. By the fundamental theorem of coalgebras, every element of A is contained in a finite-dimensional subcoalgebra. Such a subcoalgebra supplies a finite-dimensional algebraic representation of H whose matrix coefficients lie in A. The common kernel of all these representations is exactly $N(A)$: if $h \in N(A)$ and $\Delta a = \sum a_{(1)} \otimes a_{(2)}$, then

$$a(xh) = \sum a_{(1)}(x) a_{(2)}(h) = \sum a_{(1)}(x) a_{(2)}(1) = a(x).$$

The Zariski topology of H is Noetherian. Since the intersection of the kernels is $\{1\}$, finitely many of the above representations have trivial kernel intersection. Their direct sum is a faithful algebraic representation $\rho : H \to \mathrm{GL}(V)$, and every matrix coefficient of ρ belongs to A.

A faithful morphism of affine algebraic groups in characteristic zero is a closed immersion. Consequently $\mathcal{R}(H)$ is generated by the pullbacks of the coordinate functions on $\mathrm{GL}(V)$: the entries of ρ and the inverse determinant, equivalently coefficients of ρ and its dual. The former lie in A, and the latter lie in A by antipode stability. Hence $\mathcal{R}(H) \subset A$, proving equality. $\square$

5 Proof of the criterion and corollary

Proof of Theorem 1. The space $\mathcal{O}_{\exp}(G)$ is invariant under left and right translations and inversion. Restriction commutes with these operations. Therefore A_{alg} satisfies Lemma 3, and the kernel hypothesis plus Lemma 4 gives

$$A_{\mathrm{alg}} = \mathcal{R}(H).$$

In particular, the range of r contains $\mathcal{R}(H)$.

For every connected linear complex Lie group, $\mathcal{R}(H)$ is contained and dense in $\mathcal{O}_{\exp}(H)$ [3, Theorem 1]. Hence r has dense range. Corollary 3.10 of the source [2] now says that

$$\mathcal{O}_{\exp}(G) \longrightarrow \mathcal{O}(G)$$

is an Arens–Michael envelope. By source Proposition 3.6, this is equivalent to holomorphic reflexivity of $\mathcal{O}(G)$. □

Proof of Corollary 2. Put $H = G_0$ and assume first that $\widetilde{\mathrm{Lin}}_{\mathbb{C}}(G) = \{1\}$. Restriction of an exponential-type function on G is of exponential type on H: ambient word length is bounded above by a constant multiple of intrinsic word length on a compact generating set of H. For connected linearly complex reductive H, Aristov proves

$$\mathcal{O}_{\exp}(H) = \mathcal{R}(H)$$

as locally convex algebras [1, Theorem 5.9]. Thus $A_{\mathrm{alg}} = r(\mathcal{O}_{\exp}(G))$.

Let $N = N(A_{\mathrm{alg}})$. Lemma 3 shows that N is a subgroup, and every function in $\mathcal{O}_{\exp}(G)$ is constant on N. Source Corollary 3.5 then yields $N \subset \widetilde{\mathrm{Lin}}_{\mathbb{C}}(G) = \{1\}$. Theorem 1 proves holomorphic reflexivity.

Conversely, a connected linearly complex reductive group is affine algebraic and Stein. The components of G are translates of H, so the underlying complex manifold of G is Stein as well. Source Proposition 4.1 states that holomorphic reflexivity of $\mathcal{O}(G)$ for a compactly generated Stein group forces $\widetilde{\mathrm{Lin}}_{\mathbb{C}}(G) = \{1\}$. □

6 Verification and limitations

The source counterexample does not conflict with Corollary 2. Its identity component is $\mathbb{C}^\times$, which is reductive, but the source proves that every exponential-type function is constant on that component; hence its generalized linearizer is nontrivial.

The remaining obstruction is now sharp. For a general connected linear H, the regular algebra $\mathcal{R}(H)$ is dense in $\mathcal{O}_{\exp}(H)$ but may be strictly smaller. Triviality of $\widetilde{\mathrm{Lin}}_{\mathbb{C}}(G)$ makes the full restriction image separate points, but does not formally imply that its regular part does. Abstractly, on $(\mathbb{C}, +)$ the Hopf algebra generated by $e^{\pm z}$ and $e^{\pm \alpha z}$ for nonreal α has trivial common period kernel while meeting $\mathbb{C}[z]$ only in the constants. This is not an example arising from a group extension and is not a counterexample; it shows why a full proof needs additional geometry of the component extension.

The result leaves Conjecture 1.6 of the source and the general solvable-radical case of Conjecture 1.5 open.

7 Novelty check

The exact source, conjecture, generalized-linearizer terminology, and the reductive-identity-component variant were searched in the run indexes, the local parsed arXiv corpus, arXiv, and a bounded citation neighborhood through 2025. The only later citation found in that neighborhood was Aristov's HFG Hopf-algebra paper [4]; it uses the duality framework but does not state this subcase. The density note [3] supplies a dependency, not the restriction-image argument. No exact prior theorem was found. This is bounded evidence; novelty remains subject to specialist review.

References

[1] O. Yu. Aristov, *Holomorphic functions of exponential type on connected complex Lie groups*, J. Lie Theory 29 (2019), no. 4, 1045–1070; arXiv:1903.08080, Theorem 5.9.

[2] O. Yu. Aristov, *On holomorphic reflexivity conditions for complex Lie groups*, Proc. Edinb. Math. Soc. 64 (2021), no. 4, 800–821; arXiv:2002.03617.

[3] O. Yu. Aristov, *On density of polynomials in the algebra of holomorphic functions of exponential type on a linear Lie group*, Ufa Math. J. 16 (2024), no. 2, 76–80; arXiv:2304.00507, Theorem 1.

[4] O. Yu. Aristov, *Holomorphically finitely generated Hopf algebras and quantum Lie groups*, Moscow Math. J. 24 (2024), no. 2, 145–180; arXiv:2006.12175.